SUMMARY OF RESULTS ON LOCALLY QUASI-COHERENT MODULES WITH THE FLAT BASE CHANGE PROPERTY

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Allowed standard prerequisites (editorial).

- fppf-etale cohomology comparison and locality of cohomology. Sheaves on Stacks, Lemma stacks-sheaves-lemma-compare-fppf-etale; Cohomology on Sites, Lemma sites-cohomology-lemma-kill-cohomology-class-on-covering. Part (2).
- delivered companions: flat comparison and pushforward stability. delivered companions lemma-flat-comparison, lemma-pushforward-locally-quasi-coherent, lemma-check-lqc-fbc-on-covering. Parts (3)-(7).

Source reference. Original reference 076T: sources/stacks-chow/tags.txt, source lines 9178–9178. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0765: sources/stacks-chow/tags.txt, source lines 9155–9155. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 075R: sources/stacks-chow/tags.txt, source lines 9140–9140. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01FW: sources/stacks-chow/tags.txt, source lines 1885–1885. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0764: sources/stacks-chow/tags.txt, source lines 9154–9154. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06WL: sources/stacks-chow/tags.txt, source lines 8820–8820. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06WM: sources/stacks-chow/tags.txt, source lines 8821–8821. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Proposition 0.1. *Summary of results on locally quasi-coherent modules having the flat base change property.*

(1) *Let $\mathcal{X}$ be an algebraic stack. If $\mathcal{F}$ is in $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$, then $\mathcal{F}$ is a sheaf for the fppf topology, i.e., it is an object of $Mod(\mathcal{O}_\mathcal{X})$.*

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(2) *The category* $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$ *is a weak Serre subcategory of both* $Mod(\mathcal{O}_\mathcal{X})$ *and* $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$.
(3) *Pullback* f^* *along any morphism of algebraic stacks* $f : \mathcal{X} \to \mathcal{Y}$ *induces a functor* $f^* : LQCoh^{fbc}(\mathcal{O}_\mathcal{Y}) \to LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$.
(4) *If* $f : \mathcal{X} \to \mathcal{Y}$ *is a quasi-compact and quasi-separated morphism of algebraic stacks and* $\mathcal{F}$ *is an object of* $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$, *then*
   (a) *the total direct image* $Rf_*\mathcal{F}$ *and the higher direct images* $R^if_*\mathcal{F}$ *can be computed in either the étale or the fppf topology with the same result, and*
   (b) *each* $R^if_*\mathcal{F}$ *is an object of* $LQCoh^{fbc}(\mathcal{O}_\mathcal{Y})$.
(5) *The category* $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$ *has colimits and they agree with colimits in* $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$ *as well as in* $Mod(\mathcal{O}_\mathcal{X})$.
(6) *Given* $\mathcal{F}$ *and* $\mathcal{G}$ *in* $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$ *then the tensor product* $\mathcal{F} \otimes_{\mathcal{O}_\mathcal{X}} \mathcal{G}$ *is in* $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$.
(7) *Given* $\mathcal{F}$ *of finite presentation and* $\mathcal{G}$ *in* $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$ *then* $\mathcal{H}om_{\mathcal{O}_\mathcal{X}}(\mathcal{F}, \mathcal{G})$ *is in* $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$.

Proof. Part (1) is Sheaves on Stacks, Lemma 076T.

Part (2) for the embedding $LQCoh^{fbc}(\mathcal{O}_\mathcal{X}) \subset Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$ we have seen in the proof of Lemma 0765. Let us prove (2) for the embedding $LQCoh^{fbc}(\mathcal{O}_\mathcal{X}) \subset Mod(\mathcal{O}_\mathcal{X})$. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism between objects of $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$. Since $\mathrm{Ker}(\varphi)$ is the same whether computed in the étale or the fppf topology, we see that $\mathrm{Ker}(\varphi)$ is in $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$ by the étale case. On the other hand, the cokernel computed in the fppf topology is the fppf sheafification of the cokernel computed in the étale topology. However, this étale cokernel is in $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$ hence an fppf sheaf by (1) and we see that the cokernel is in $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$. Finally, suppose that

$$0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$$

is an exact sequence in $Mod(\mathcal{O}_\mathcal{X})$ (i.e., using the fppf topology) with $\mathcal{F}_1$, $\mathcal{F}_2$ in $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$. In order to show that $\mathcal{F}_2$ is an object of $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$ it suffices to show that the sequence is also exact in the étale topology. To do this it suffices to show that any element of $H^1_{fppf}(x, \mathcal{F}_1)$ becomes zero on the members of an étale covering of x (for any object x of $\mathcal{X}$). This is true because $H^1_{fppf}(x, \mathcal{F}_1) = H^1_{étale}(x, \mathcal{F}_1)$ by Sheaves on Stacks, Lemma 075R and because of locality of cohomology, see Cohomology on Sites, Lemma 01FW. This proves (2).

Part (3) follows from Lemma 0764 and Sheaves on Stacks, Lemma 06WL.

Part (4)(b) for $R^if_*\mathcal{F}$ computed in the étale cohomology follows from Lemma 0765. Whereupon part (4)(a) follows from Sheaves on Stacks, Lemma 075R combined with (1) above.

Part (5) for the étale topology follows from Sheaves on Stacks, Lemma 06WM and Lemma 0764. The fppf version then follows as the colimit in the étale topology is already an fppf sheaf by part (1).

Parts (6) and (7) follow from the corresponding parts of Lemma 0764 and Sheaves on Stacks, Lemma 06WM. □